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V12C

Virudhunagar District
Common Haify Yearly Examination, - 2025

Standard 12

CHEMISTRY

Time: 3.00 Hours Marks: 70

PART - I

I. Choose the correct answer and answer all the questions: 15x1=15

1) Considering Ellingham diagram, which of the following metals can be used to reduce alumina?
a) Fe b) Cu c) Mg d) Zn

2) Carbon atoms in fullerene with formula C_{60} have
a) sp^3 hybridised
b) sp hybridised
c) sp^2 hybridised
d) partially sp^2 and partially sp^3 hybridised

3) The molarity of given orthophosphoric acid solution is 2M, its normality is
a) 6 N b) 4N 2) 2N d) none of these

4) Which of the following oxidation states is most common among the lanthanoids?
a) 4 b) 2 c) 5 d) 3

5) What is the coordination number of the following complex $Na_2[Ni(EDTA)]$?
a) 2 b) 4 c) 6 d) 1

6) CsCl has bcc arrangement, its unit cell edge length is 400 Pm, its inter atomic distance
a) 400 Pm b) 800 Pm c) $\sqrt{3} \times 100$ Pm d) $\frac{\sqrt{3}}{2} \times 400$ Pm

7) During the decomposition of H_2O_2 to give dioxygen, 48 g O_2 is formed per minute at certain point of time. The rate of formation of water at this point is
a) 0.75 mol min^{-1} b) 1.5 mol min^{-1} c) 2.25 mol min^{-1} d) 3.0 mol min^{-1}

8) The pH of 10^{-5} M KOH solution will be
a) 9 b) 5 c) 19 d) none of these

9) Faradays constant is defined as
a) charge carried by 1 electron b) charge carried by one mole of electrons
c) charge required to deposit one mole of substance
d) charge carried by 6.22×10^{10} electrons

10) Which one of the following compounds used as eye lotion?
a) Argyrol b) colloidal calcium c) milk of magnesia d) streptomycin

11) Williamson synthesis of preparing dimethyl ether is a /an/
a) SN^1 reactions b) SN^2 reactions
c) electrophilic addition d) electrophilic substitution

12) In which of the following reactions new carbon-carbon bond is not formed?
a) Aldol condensation b) Friedel Craft reaction
c) Kolbe's reaction d) Wolf Kishner reduction

13) Ammonium salt of benzoic acid is heated strongly with P_2O_5 and the product so formed is reduced and then treated with $NaNO_2$/HCl at low temperature. The final compound formed is
a) Benzene diazonium chloride b) Benzyl alcohol
c) Phenol d) Nitrosobenzene

14) Complete hydrolysis of cellulose gives
a) L-Glucose b) D-Fructos c) D-Ribose d) D-Glucose

15) Which one of the following is a bio-degradable polymer?
a) HDPE b) PVC c) Nylon-6 b) PHBV

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PART - II

II. Answer any six questions. Q.No.24 is compulsory: **6x2=12**

16) Mention any two uses of Zinc.
17) What is inert pair effect?
18) Give an example of co-ordination compound used in medicine and two examples of biological important co-ordination compounds.
19) Write Arrhenius equation and explains the terms involved.
20) Define: Electro chemical equivalent.
21) Explain Dows process.
22) Write short notes on Levine and Hauser acetylation reaction.
23) What are food preservatives?
24) Write the expression for the solubility product of Hg_2Cl_2

PART - III

III. Answer any six questions. Q.No. 33 is compulsory: **6x3=18**

25) Write a note on metallic nature of P-block elements.
26) What are interstitial compounds? Give two properties.
27) Give any three characteristics of ionic crystals.
28) Derive an expression for Henderson equation.
29) What is the difference between a sol and gel?
30) Write note on reducing property of Formic acid.
31) Give any three difference between DNA and RNA.
32) How is terylene prepared?
33) Predict the products A, B and C in the following sequence of reaction:

$$C_6H_6 + \text{Propene} \xrightarrow[523K]{H_3PO_4} A \xrightarrow[5\%\ aqu\ Na_2CO_3]{Air\ O_2} B \xrightarrow{H_2SO_4} C + \text{acetone}$$

PART - IV

IV. Answer all the questions: **5x5=25**

34) a] Explain the extraction of Aluminium by using Hall-Herold Process.

(OR)

b] i) Describe the structure of diborance
ii) Give the two uses of sulphuric acid.

35) a] Write the postulates of Valence Bond Theory.

(OR)

b] i) Write the electronic configuration of Ce^{4+} and CO^{2+}
ii) Write short note on metal excess and metal deficiency defect with an example.

36) a] Explain the function of H_2–O_2 fuel cell.

(OR)

b] i) What is an elementary reaction? Give two difference between order and molecularity of a reaction
ii) Define ionic product of water.

37) a] Explain Intermediate compound Formation theory of catalysis with an example.

(OR)

b] Explain the mechanism of Cannizaro reaction.

38) a] Explain the structural elucidation of Glucose.

(OR)

b] i) Write short notes on Mustard oil reaction.
ii) Write a note on Vulcanization of rubber.

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